

AI Applications Portfolio

Melvis Maduagwu

A comprehensive showcase of coursework, case studies, and industry research across 10+ domains of artificial intelligence application.



About This Portfolio



This portfolio represents a semester of deep exploration into how artificial intelligence is transforming real-world industries. Each module combined theoretical concepts with practical case studies, producing a body of work that spans from healthcare to blockchain, retail to smart cities.

12+

Research
Documents

10+

Industry
Domains

1

In-Depth
Case Study

Domains covered: Finance · Healthcare · Agriculture · Fraud Detection · Blockchain · Entertainment · Government · Manufacturing · Retail · Smart Cities · Ethics · Societal Impact

Coursework Overview

10 domains explored across theory, application, and societal impact



Finance



Healthcare



Fraud Detection



Blockchain



Entertainment



Government



Manufacturing



Retail



Smart Cities



AI & Ethics

Key Learning: Finance & Fraud Detection

JPMorgan Chase Case Study

Analyzed how JPMorgan deployed OmniAI + JADE to build a multi-layer AI fraud detection pipeline.

Studied NLP for Business Email Compromise, graph analytics for Anti-Money Laundering, and behavioral profiling for real-time risk scoring.

AI Use Cases in Finance:

Algorithmic trading · Credit scoring · Portfolio optimization
AML detection · Customer chatbots · Insurance underwriting

\$1.5B

Fraud losses prevented annually

300×

Faster fraud detection vs. legacy systems

95%

Reduction in AML false positives

Key Learning: Healthcare & AGI

✓ Solved

- Medical image analysis (chest X-rays, mammograms)
- Diabetic retinopathy screening
- Stroke detection via CT/MRI
- Remote patient monitoring via wearables

⚡ In Progress

- Personalized medication recommendations
- Drug discovery timeline compression
- Predictive deterioration alerts in ICUs

🔬 Not Yet Feasible

- Equitable healthcare across demographics
- Explainable AI for clinical decisions
- Clear legal accountability frameworks
- HIPAA/GDPR compliant AI training

AGI Insight: Unlike narrow AI (pattern matching on existing data), AGI could form hypotheses, run experiments, and discover cures for rare diseases autonomously — but brings serious alignment and bioweapon risks.

Key Learning: Ethics, Society & AI in Hiring



Bias amplification at scale — AI inherits & automates historical hiring discrimination



Black-box decisions — candidates cannot contest AI-driven rejections



Privacy invasion — video analysis, psychometrics & behavioral data collected without meaningful consent



Regulatory gap — NYC bias audit law & EU AI Act emerging but enforcement still patchy

AI Hiring Use Cases

- Resume screening & ATS
- Job description optimization
- Candidate sourcing (LinkedIn)
- Chatbot screening
- Video interview analysis
- Skills assessments
- Predictive hiring analytics
- Internal mobility tools

Key Learning: Smart Cities · Manufacturing · Retail



Smart Cities

- › Traffic optimization (Hangzhou: –15% travel time)
- › Predictive maintenance for water/power infrastructure
- › AI waste routing using Radio Frequency Identification bins (Seoul: –40% truck trips)
- › Crime prediction, smart parking, urban planning



Manufacturing

- › Predictive maintenance → 30–50% less downtime
- › Computer vision quality control (50% fewer defects)
- › Digital twins for real-time simulation
- › Energy optimization saving 10–20% on costs



Retail

- › Dynamic pricing & real-time demand forecasting
- › Amazon cashierless Go stores using CV + sensors
- › Visual search & virtual try-on (IKEA, ASOS)
- › AI supply chain resilience during disruptions

Potential Projects

Ideas directly inspired by coursework & ready to build

01

AI Fraud Detection Dashboard

Visualize real-time transaction risk scoring using ML models. Inspired by JPMorgan's OmniAI pipeline.

Finance · ML · Data Viz

02

AI Hiring Bias Auditor

Tool that ingests a company's ATS scoring data and flags demographic disparities, with explainability reports.

Ethics · NLP · Fairness AI

03

Smart City Resource Optimizer

Simulate traffic signal timing or waste collection routing using reinforcement learning and live sensor data.

Smart Cities · RL · IoT

04

Healthcare Diagnostic Assistant

A proof-of-concept image classifier for medical scans, exploring the solved vs. unsolved problem space in AI diagnostics.

Healthcare · Computer Vision · CNN

Cross-Cutting Themes & Skills Gained

Bias & Fairness

Every domain — from healthcare to hiring — revealed that AI trained on historical data systematically amplifies existing inequities.

Explainability vs. Performance

Black-box models achieve better accuracy but fail regulatory and trust requirements. XAI is a non-negotiable design consideration.

Human-AI Collaboration

The most successful deployments augment human judgment rather than replacing it — Data analysts, physicians, fraud analysts all remain essential.

Regulatory Lag

Technology consistently outpaces governance. The EU AI Act, NYC bias laws, and GDPR/HIPAA are early but incomplete frameworks.

Skills Developed

- ✓ Industry AI research & analysis
- ✓ Case study methodology
- ✓ Ethical AI frameworks
- ✓ Problem categorization (solved / partial / infeasible)
- ✓ Cross-domain pattern recognition
- ✓ Technical writing & synthesis
- ✓ Stakeholder impact assessment



Thank You

for reviewing my portfolio

This portfolio has helped me to demonstrate my ability to analyze, critique, and contextualize AI across industries; combining technical understanding with ethical and societal awareness.



Resources

<https://qa-financial.com/inside-jpmorgans-18bn-qa-push-with-omniai-reshaping-testing/>

<https://www.constellationr.com/insights/news/jpmorgan-chase-digital-transformation-ai-and-data-strategy-sets-generative-ai>

<https://pmc.ncbi.nlm.nih.gov/articles/PMC9636573/>

<https://www.retina-specialist.com/article/ai-for-dr-screening-where-are-we-in-2025>

<https://www.theguardian.com/environment/2025/dec/18/smart-bins-measuring-food-waste-south-korea>